

Calibrate Before You Rank: Corruption-Validated Quality Metrics for Rhythm-Game Charts

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Abstract

Rhythm-game charts are designed artifacts with many valid realizations per song, yet generated charts are still evaluated with metrics borrowed from other tasks: F-measure against one reference chart, language-model (LM) perplexity, and unreferenced statistics such as entropy, distinct- n , and self-similarity. We validate metrics by measuring their response to controlled damage. Applying nine targeted corruption operators at increasing doses to 170 human charts (4,760 scored charts including the originals), we find failures that spot checks alone do not reveal: perplexity *improves* by 37% when charts are flattened toward the LM’s modal patterns, self-similarity structureness *rises* by 62% under copy-paste collapse, and a global 60 ms audio misalignment—ten times the perceptual threshold—goes unregistered not only by all 56 chart-only quantities we compute (a structural translation invariance; maximum response 0.03 standard deviations) but also by the standard matching-based offset statistic. A two-layer protocol repairs these failures: raw diagnostics are normalized against per-difficulty target bands calibrated on the official-chart corpus, and a perceptually anchored timing family—a 6 ms just-noticeable-difference deadzone, severity tiers at its multiples, and a fixed-lattice phase detector over a two-tier grid-reference hierarchy—recovers injected 15/30/60 ms shifts with exact medians, on at least 99.4% of charts per grid tier and dose, without a reference chart. The corruptions also establish complementarity: a sparse-outlier corruption that grid matching largely absorbs is detected at the token level by the grammar family, while the anchor shift that no chart-only metric registers is detected only by the timing family—the score profile cannot be reduced to a scalar. Applied to five published generators across three chart vocabularies, the profile replaces single-number rankings with mechanism-level diagnoses. We release the corruption operators, calibration artifacts, per-chart records, and an installable evaluation package. [PENDING: P8: artifact URL.]

1 Introduction

Automatic chart generation for rhythm games—mapping audio to a playable sequence of timed, typed notes—has progressed from onset-detection pipelines to conditional sequence models [1–4]. Evaluation has not kept pace. Because a chart is a designed artifact rather than an an-

notation, the same song admits many valid charts, and the field’s dominant tool—F-measure against the single official chart—penalizes every valid alternative equally and cannot be computed for songs without charts. The usual complement is reference-free measurement, for which authors borrow from symbolic music and text generation: LM perplexity [1], pattern-space congruence [2], event entropy, distinct- n [5], Self-BLEU [6], feature-distribution comparison [7, 8], and self-similarity structureness [9].

This paper asks whether those borrowed metrics measure chart quality, and answers with a validation method that requires no listening test: *targeted corruption*. Starting from human (official) charts, we apply nine damage operators—dense timing jitter, sparse severe outliers, global anchor shift, type shuffling, loop collapse, LM-argmax flattening, density scaling, burst insertion, bar shuffling—at three doses each. Corrupting a human chart does not, in expectation, improve it, so a usable quality metric must respond in the correct direction, and the response must grow with dose. Metric validation becomes an automated, reproducible procedure that requires neither human raters nor model training, and applies unchanged to any proposed chart metric.

The audit yields a negative result with a constructive core. Borrowed metrics fail structurally (Section 3.2): perplexity rewards the flattening it should penalize, structureness mistakes degenerate repetition for form, several statistics are near-proxies for chart length, and every chart-only quantity is insensitive to global audio misalignment. The underlying raw quantities carry signal; what is missing is calibration and grounding. Normalizing each diagnostic against per-difficulty target bands estimated from the official corpus restores a consistent, manipulation-resistant reading (Section 2.2, Section 3.2), and a timing family anchored on a grid-reference hierarchy—authored per-bar timing metadata when available, an audio-estimated beat grid otherwise, never a generated chart’s self-declared grid—closes the misalignment gap without a reference chart (Section 2.3, Section 3.3).

Relation to prior work. No metric in the chart-generation literature has been audited against controlled damage. Dance Dance Convolution evaluates step placement by F-measure against a reference chart and step selection by LM perplexity [1]; later systems inherit this template [3, 4], and TaikoNation adds pattern-congruence measures against human charts [2]. In generative-music evaluation, Yang and Lerch compare feature distributions between generated and reference sets [7], MusPy popu-

larized symbolic statistics [8], and the Jazz Transformer introduced self-similarity structureness [9]; these are set-level or direction-ambiguous quantities, and we audit their behavior when used as per-chart quality signals. Theis et al. showed that likelihood and sample quality can diverge [10]; our flattening result is a corpus-LM instance of that dissociation with a concrete attack. Unlike pure critiques, we pair each failure with a repair; unlike human-anchored metric studies, the harness requires no raters. Onset F-measure follows `mir_eval` [11]; timing thresholds follow the psychophysics of temporal discrimination [12, 13] rather than any game’s scoring windows.

Our contributions are:

1. **A corruption-based audit harness:** nine dose-controlled damage operators with acceptance criteria (population-level monotone response, resistance to manipulation, published coupling structure) and pre-registered fairness rules, applicable unchanged to any future chart metric (Section 2.4).
2. **A structural failure analysis** of borrowed metrics: two sign-reversal results (perplexity improves under flattening; structureness improves under collapse), an insensitivity result covering all 56 chart-only quantities under anchor shift, and a direction-instability result—perplexity ranks correctly against systems on the high-entropy side of the corpus yet inverts on the low-entropy side, shown constructively by C5 (Section 3.2).
3. **A two-layer, reference-free protocol** with an empirically assigned role taxonomy and a perceptually anchored timing family: one 6 ms just-noticeable-difference constant grounds a deadzone, severity tiers, and a clean-note rate; a piecewise fixed-lattice phase detector recovers injected shifts where matching statistics do not respond (Sections 2.2–2.3 and 3.3).
4. **Complementarity evidence:** three corruptions, each undetected by one metric family and strongly detected by another, in both directions—the empirical case that chart quality is a profile, not a scalar (Section 3.4).
5. **Application to five published systems** across three chart vocabularies, yielding mechanism-level diagnoses and an applicability matrix instead of a single ranking (Section 3.5), plus released artifacts: an installable package, corruption operators, calibration bands, per-chart records, and grouped song-level dataset splits. [\[PENDING: P8: public repo URL.\]](#)

The corruptions establish what metrics detect: *damage*. Whether the surviving metrics also track player experience is a distinct question that requires perception studies; we return to it in Section 4.

2 Methods

2.1 Data and grid references

We use a corpus of 1,155 taiko-style chart files covering 1,013 unique songs. The files are community transcrip-

tions of each song’s official difficulty courses; we call them official charts and use them as the human reference corpus. Songs are grouped by audio identity and normalized title and split at the song level into 813/80/120 training/validation/test song groups, so that no song appears in more than one split. Calibration uses the official difficulty-course charts of the training songs; evaluation uses 40 held-out test songs (170 official course charts) across all difficulty courses.

Timing is the one family that cannot be measured from the chart alone: it needs a grid reference. We fix a two-tier hierarchy: (1) the *authored* grid—per-bar timing metadata (bar lines, per-bar meter) shipped with the song’s official charts, which contains no note positions and is therefore usable in reference-free evaluation; (2) an *estimated* grid from an audio beat tracker, for songs without authored metadata. A generated chart’s self-declared grid is never consulted: a system that supplied its own reference could trivially zero its own offsets. On our corpus the estimated tier is accurate: estimated beat periods agree with authored tempo within 2% on 40/40 test songs. Period agreement does not bound downbeat phase; the shift recovery reported below is phase-invariant, and absolute offsets are reported on the authored tier.

2.2 Two-layer scoring

Layer 1: raw diagnostics. Each chart is a sequence of timed, typed events. We compute quantities with native units and no claimed direction: note density and local-density percentiles, inter-onset-interval (IOI) entropy, color-switch rate, n -gram LM negative log-likelihood (trigram, add- α smoothing, trained on the calibration corpus), unseen-transition rate, unique/repeated 4-gram rates, and windowed LM surprisal percentiles. These deliberately coincide with the borrowed metrics: the audit compares *normalization strategies*, not measurement choices. Score names used later map onto these directly: rhythm complexity is the band score over IOI entropy; repetition and variety adequacy over the repeated and unique 4-gram rates; density adequacy and the spike score over density and local-density percentiles; transition validity over the unseen-transition rate; and pattern information content (pattern-IC) over the windowed LM surprisal percentiles.

Layer 2: calibrated band scores. For each diagnostic and difficulty course we estimate a target band $[p_{10}, p_{90}]$ over the calibration corpus. Scores are 1 inside the band and decay rationally outside $(1/(1 + d^2))$ in units d of half the band width; error-type quantities use exponential penalties. The rational tail is calibrated to the data: held-out official charts reach up to 4.9 band units on the language axis (creative outliers), and a Gaussian tail would collapse that entire range together with far-pathological readings to 0.00. Each band score is accompanied by a signed band-position diagnostic in the same units, which does not saturate. Band scores measure *typicality*: “more entropy” is neither good nor bad, but “entropy within the range human charters use at this difficulty” is well defined, and the score is reference-free at the song level—scoring a chart for a new song requires only the fixed corpus calibration.

Two metric types. The suite separates *defect-type* quantities, whose direction is unconditional (transitions that never occur in the human corpus; violations of playability constraints), from *typicality-type* quantities, where only the corruption evidence, not the sign of a deviation, justifies an interpretation. For typicality scores the inference is one-directional: the corruptions certify that damage drives charts out of band; they do not certify the converse. An out-of-band reading is a flag whose meaning is adjudicated by the rest of the profile (Section 4). Because the corruptions start from corpus charts, some sensitivity of typicality scores is by construction; the certified properties that a bare distance-to-corpus statistic would not deliver are boundedness under the manipulation attack, the insensitivity results, and the coupling structure (Section 3.2–3.4).

Role taxonomy. Scores are assigned roles empirically by the audit, not a priori: *rankers* respond to damage and separate systems (pattern-IC, transition validity, repetition and variety adequacy, rhythm complexity); *constraints* saturate near 1 for all reasonable charts—on the held-out official charts, 86–98% score at least 0.99 on each constraint, median 1.0—flag only gross violations (density, strain, overload, spike, and an aggregate playability gate), and act as vetoes—they never average into a score; *diagnostics* carry signal with direction-only interpretation (the fixed-lattice phase detector; the hypersensitive rare- n -gram rate). We report score profiles, not a single aggregate: fixing trade-off weights between axes is exactly the judgment we defer to human data. Table 1 summarizes the resulting five-plus-one metric families.

2.3 Perceptually anchored timing family

All timing thresholds are perceptual, not game-specific: a single 6 ms deadzone—the just-noticeable timing distortion for short intervals [12]—below which deviations are treated as imperceptible; severity tiers at $1 \times / 2 \times / 3 \times$ the deadzone, spanning from this conservative attentive-listening floor up to magnitudes relevant in play; and a relative channel that compares consecutive inter-onset intervals against a Weber envelope $\max(6 \text{ ms}, 2.5\% \times \text{IOI})$, computed on raw offsets before any deadzone is applied. The family reports a partition of all notes—*clean* (matched within the deadzone), tier exceedances, and unmatched notes—so that the primary statistic, the *clean rate*, has unmatched notes in its denominator and cannot be improved by dropping notes. Aggregation is per-chart first, never pooled over notes: a chart is judged by its own worst notes, and cross-chart tails are reported alongside medians.

Global misalignment needs a dedicated detector. A matching-based signed offset (match each note to its nearest lattice anchor, average the residuals) re-matches every shifted note on a dense subdivision lattice and therefore cannot register a constant shift (Section 3.3). The *fixed-lattice phase detector* instead finds the global offset that maximizes support of the note stream on a lattice laid inside each authored bar (eighth-note, beat, and bar-line scales combined), searched over $\pm 250 \text{ ms}$ with minimum-norm tie-breaking. Because the lattice follows the authored bar lines, it is aperiodic under tempo changes and remains well

defined on songs with meter and tempo variation.

2.4 Corruption operators and acceptance criteria

Nine operators each damage one chart property at three controlled doses (Table 2); they are chosen to span the family taxonomy of Table 1, one to two per family. Band scores are two-sided, so a corruption can push an individual off-band chart *toward* its band; direction is therefore guaranteed only at the population level, and all criteria are evaluated over the 170 charts. The certification statistic is the per-chart *net direction*: the fraction of charts whose value rises minus the fraction whose value falls, each chart paired with its own intact version. With 170 charts its null width is roughly $\pm 2/\sqrt{170} \approx \pm 0.15$; every detection claimed below exceeds this by at least a factor of five, and no cell-level significance is claimed for the published response matrix. Dose–response correlations summarize sensitivity where magnitudes are informative; net direction is used where re-matching effects saturate magnitudes. Operators that directly edit a measured quantity (C6 on density, C7 on local density) make their diagonal response a manipulation check rather than evidence; the informative cells are the off-diagonal responses and the insensitivities. Acceptance requires (a) *sensitivity*: the designated target degrades with dose; (b) *resistance to manipulation*: no score family rewards damage without bound; and (c) *disclosed coupling*: the full operator $\times$ metric response matrix is published rather than summarized by its diagonal.

Fairness rules fixed in advance: identical corruptions for baselines and calibrated scores; identity pairs across layers (distinct-4 $\equiv$ unique-4-gram rate; transition validity $\leftrightarrow$ unseen-transition rate) are never counted as independent agreement; length confounds are checked against note count; and the flattening attack is additionally run *cross-LM*—driven by an LM trained on half the calibration songs and measured by an LM trained on all of them—so that perplexity’s failure cannot be an artifact of attacking its own model.

2.5 Evaluated systems

We evaluate five published generators spanning three chart vocabularies: two taiko-domain systems—Mapperatorinator [14], an open-source osu!-ecosystem generator run in taiko mode [PENDING: E: *positioning wording*], and TaikoNation [2]—plus, for the game-agnostic families, Dance Dance Convolution’s onset pipeline (ddconset) [1], the public timing-only configuration of GenéLive! [3], and an osu! difficulty-targeted generator (autoosu) [15]. The three vocabularies are taiko note types, timing-only onset streams, and osu! position-based objects. All systems chart the same 40 held-out songs from audio in their public inference configurations; none was trained on this corpus, although overlap between these commercial songs and the systems’ own training data cannot be excluded (Section 4). Difficulty mappings from each system’s controls onto the five courses were fixed before evaluation and ship with the artifacts. Format conversion into the evaluation event stream is shared within each vo-

Family	Measures	Inputs	Type	Tier
timing	placement vs. grid	times + grid	defect	diag.+rank
grammar	sequence idiom	tokens + LM	both	ranker
structure	learnable organization	tokens	typicality	ranker
coupling	chart–music relation	times + audio	typicality	ranker
gap	distance to human corpus	features	typicality	ranker
constraints	playability limits	tokens	defect	veto

Table 1: The five-plus-one metric families. *Inputs* determine cross-vocabulary applicability (Section 3.5): timing and coupling read note times only and apply to any timed note stream given a grid reference (the constraint quantities are likewise time-based, though their gates are calibrated to taiko course norms); grammar, structure, and gap require a vocabulary tokenizer and a calibration corpus. Perceptual constants are fixed from the temporal-discrimination literature [12, 13]: a 6 ms absolute deadzone equal to the relative-distortion floor, a 2.5% Weber fraction of the local inter-onset interval, severity tiers at $1 \times / 2 \times / 3 \times$ the deadzone. LM: trigram, $\alpha=0.05$; bands: per-course $[p_{10}, p_{90}]$.

Operator (dose parameter)	Damages	Designated target
C1 timing jitter ($\pm 10/20/30$ ms)	micro-timing	rhythm cmplx. + timing
C1s sparse outliers ($0.5/1/2\%$, ± 60 ms)	timing tail	(discovered: grammar)
C2 anchor shift ($+15/30/60$ ms)	audio alignment	timing detector
C3 type shuffle ($p=0.2/0.4/0.8$)	note grammar	transition validity
C4 loop collapse ($p=0.3/0.6/1$)	structure	repetition adequacy
C5 flattening ($p=0.3/0.6/1$)	interestingness	(manipulation test)
C6 density scale ($\times 0.5/1.5/2$)	difficulty fit	density adequacy
C7 burst insert ($2/4/8$ clusters)	playability	density-spike score
C8 bar shuffle ($p=0.3/0.6/1$)	global form	repetition adequacy

Table 2: Nine corruption operators applied to 170 official charts at three doses each (4,760 scored variants including originals; deterministic seeds; operator outputs re-verified bit-identically by a separate re-execution script against the archived records). Every operator preserves what it does not target: C3–C5 preserve all note times exactly; C4 and C6 preserve or control density. C1s displaces $\max(1, \lceil np \rceil)$ of a chart’s n notes by ± 60 ms ($10 \times$ the deadzone): a few severely misplaced notes in an otherwise intact chart, the failure mode that mean statistics cannot register.

cabulary; the two systems sharing the osu! conversion path read different phase offsets below (19 vs. 26 ms), which rules out a shared conversion constant, though per-format offsets cannot be fully excluded (Section 4). Official charts of the same songs provide the reference row. Systems that condition on difficulty produce one chart per official course (170 or 200 charts); the others produce one per song (40).

3 Results

3.1 Sensitivity: targeted damage moves the right scores

All six operators with calibrated band targets pass: dose–response Spearman $\rho(\text{dose}, \text{target score})$ is negative for C1 rhythm complexity (−0.48), C3 transition validity (−0.33), C4 repetition adequacy (−0.41), C6 density adequacy (−0.68), C7 density spike (−0.79), and C8 repetition adequacy (−0.48), with monotone mean trajectories (all $p < 10^{-18}$; the unit of analysis is the chart–dose pair, so these p values summarize strength, not independent evidence). The same six targets have per-chart net direction −0.55 to −0.95 at maximum dose. The two families whose inputs the chart-only run cannot cover are audited in the same way: the audio-coupled at-onset statistic detects the anchor shift (net direction −0.74 to −0.88 across C2 doses; median envelope z $1.09 \rightarrow -0.34$ at 60 ms) while staying

exactly at zero under time-preserving flattening, and the manifold-gap score responds to every token-editing operator (net direction −0.19 to −0.81 at maximum dose) while being translation-invariant under C2 (−0.05), as any chart-only quantity must be. C2’s target is the timing detector (Section 3.3); C1s’s response pattern is the complementarity result (Section 3.4); C5’s bounded response is the manipulation-resistance property, reported next.

3.2 Borrowed metrics reward damage

Two borrowed metrics reverse sign under damage (Figure 1, left). *Perplexity rewards flattening*: throughout, perplexity is the exponential of a chart’s mean per-note trigram NLL, averaged over charts. Official test charts average 9.44 under the corpus trigram; rewriting them toward the LM’s modal patterns lowers perplexity by 37% ($9.44 \rightarrow 5.98$). Dance Dance Convolution reports perplexity as model fit on held-out human charts [1]; transplanted to generated output as a reference-free quality reading—a tempting transfer given its correct direction on real systems below—the lower-is-better reading prefers the damaged chart—a corpus-LM instance of the likelihood/quality dissociation [10]—and the reversal survives the cross-LM version of the attack ($9.44 \rightarrow 6.04$, −36%). The band score over the same statistic saturates at its ceiling of 1—flattening cannot push a chart above typicality—and the damage the attack actually does is registered where it lands: repetition adequacy falls from

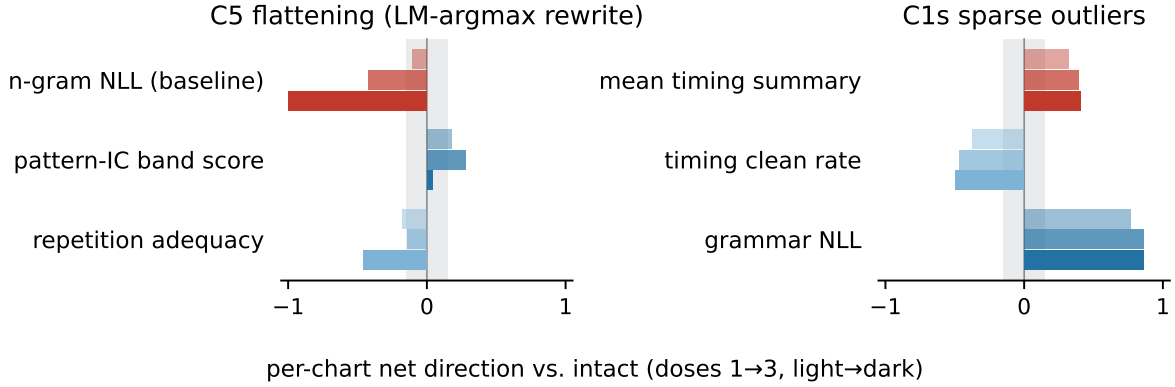


Figure 1: Two corruptions that one metric family does not register and another detects. Bars show per-chart net direction (fraction of charts moving up minus moving down relative to their own intact version) at doses 1–3, shaded light to dark; the grey band is the null width $\approx \pm 0.15$. **Left (C5 flattening):** rewriting notes toward the LM argmax drives n -gram NLL down in every chart (-1.0 at full dose)—an improvement under the lower-is-better reading—while the band score over the same statistic shows no dose-consistent reward ($+0.18/+0.28/+0.04$, bounded at typicality) and repetition adequacy registers the induced looping. **Right (C1s sparse outliers):** displacing 0.5–2% of notes by ± 60 ms leaves mean timing summaries with zero median response (their positive net direction reflects sub-resolution drift with no usable magnitude), moves the timing clean rate in 34–45% of charts (the subdivision lattice re-matches $\approx 70\%$ of displaced notes within the deadzone), but flips the IOI token class of adjacent intervals: grammar NLL responds at $+0.77$ with one to three displaced notes per chart, rising to $+0.87$.

0.99 to 0.87. *Structureless rewards collapse:* replacing charts with their own dominant 4-gram loop raises self-similarity structureness [9] by 62% ($0.0848 \rightarrow 0.1370$)—copy-paste output is maximally self-similar—and set-level Self-BLEU rises from 0.845 to 0.898, while repetition and variety adequacy fall ($0.99 \rightarrow 0.83$ and $0.99 \rightarrow 0.90$).

Two observational results complete the case. *Direction instability:* against the two published taiko generators of Section 3.5, whose outputs lie on the high-entropy side of the human corpus, perplexity separates official from generated charts in the correct direction (AUC 0.80 on the two systems available)—yet C5 proves the same reading inverts on the low-entropy side. A signed quality metric whose sign depends on which side of the human distribution the system sits cannot be used for ranking; band scores are stable on both sides by construction. *Length confounds:* on the same pool, type–token ratio correlates with note count at $|\rho| = 0.85$ and distinct-2 at 0.56, against at most 0.28 for the calibrated rankers.

3.3 Anchor shift: insensitive by construction, undetected in practice

A chart can be idiomatic on every chart-only axis and still be displaced from the music entirely—C2 makes this concrete. That the 56 chart-only quantities do not register the shift is structural, not empirical: chart-only statistics are translation-invariant up to window-edge effects (measured maximum response 0.03 standard deviations; the 56 comprise all Layer-1 diagnostics, their band scores, and the borrowed baselines, enumerated in the released records). The finding is that the field evaluates almost exclusively with such quantities, and that the timing-aware repair fails as well: a matching-based signed offset does not respond,

for the reason given in Section 2.3—a dense subdivision lattice re-matches every shifted note. The fixed-lattice phase detector recovers the injected shifts with exact medians (15.00/30.00/60.00 ms) on both grid tiers, on at least 99.4% of charts (169 or 170 of 170) per tier and dose. Under dense jitter (C1) the rest of the timing family also certifies cleanly: the clean rate falls in 100% of charts at every dose, and the first-tier exceedance at ± 10 ms jitter matches its theoretical value ($P(|U(-10, 10)| > 6) = 0.4$; measured median 0.40).

3.4 Complementarity: the profile cannot be reduced

Three corruptions produce three different detection dissociations, each readable directly off the published response matrix (Figure 1, Figure 2):

- **C2 anchor shift:** registered by none of the 56 chart-only quantities and not by matching-based offsets; detected by the timing-family phase detector and by the audio-coupled at-onset statistic (Section 3.1).
- **C1s sparse outliers:** largely absorbed by lattice matching ($\approx 70\%$ of displaced notes re-match within the deadzone; mean summaries show exactly zero median response); detected by the grammar family (net direction $+0.77$ to $+0.87$ across doses), because a 60 ms displacement necessarily flips the IOI token class of two adjacent intervals. Density does not respond ($|\text{net direction}| \leq 0.01$); structure responds weakly (-0.12 at maximum dose, an order of magnitude below the grammar response)—the response is concentrated, only not where the operator was aimed.
- **C5 flattening:** rewards the grammar baseline (per-

plexity), is bounded by the band layer, and is detected by the structure family.

Each family has a corruption it cannot register and a corruption only it registers. Any scalar aggregate would have to fix exchange rates between axes that these results show are not exchangeable.

3.5 System profiles

Figure 3 and Figure 4 give the profiles of the five systems of Section 2.5; three findings illustrate what the profile adds over single numbers.

Equal scores, different failure mechanisms. The two taiko systems have nearly equal timing clean rates (medians 0.59 vs. 0.58; indistinguishable on this axis), yet opposite mechanisms. Mapperatorinator is internally the more coherent of the two (relative-channel violations 0.10, a quarter of TaikoNation’s 0.41; officials: 0.00) but phase-misanchored: the phase detector reads a 26 ms median offset, while the matching-based statistic reads under 1 ms—the same re-matching failure that C2 demonstrated, measured on a published system. TaikoNation deviates on both channels (relative violations 0.41; phase offset 55 ms) and, uniquely among the five, shows no relation to the audio’s dynamics: its density–energy correlation is -0.003 and its notes land on average-energy frames (at-onset envelope $z = -0.004$, against 1.09 for officials). The onset pipeline anchors accurately (2.5 ms), which doubles as a control: all five systems charted the same audio files, so the 26–55 ms offsets are not audio-decoding artifacts (per-format conversion offsets are bounded by the control in Section 2.5).

Complementary deviations. On chart-vocabulary families the two taiko systems deviate on complementary axes (Figure 4). Mapperatorinator writes in-band repetition structure (repeated-4-gram rate 0.60 on oni against the oni band 0.45–0.72; repetition and variety adequacy 1.0) but its difficulty conditioning does not transfer: its oni and ura courses sit inside their density bands while its easy/normal/hard course medians run 1.5–1.8 $\times$ over their bands’ upper edges (easy: 2.9 notes/s against a band of 0.8–1.8). This is a deviation against the requested course’s band that also trips the playability gate—under the adjudication of Section 4, a defect, not novelty—and only per-difficulty calibration can register it. TaikoNation shows the reverse pattern: density and strain in band, but grammar far outside the corpus (mean perplexity 158 against 9.4 for officials; 55% of transitions never occur in the human corpus; repeated-4-gram rate 0.004, i.e. no learnable repetition). Training on a different charting community’s corpus may account for part of that magnitude; the defect-type reading (unseen transitions) and the absent audio coupling place it on the damage arm of the adjudication regardless.

Progress statements need profiles. Between TaikoNation (2021) and Mapperatorinator (2024) the timing clean rate is unchanged—a single-number comparison would report three years without progress. The profile shows where the progress went: relative timing coherence improved by a factor of four (violations 0.41 $\rightarrow$ 0.10) and repetition structure moved from absent (repeated-4-gram rate 0.004) to

in-band (0.60 on oni), while the grammar axes, anchoring, and difficulty calibration remain open problems.

4 Discussion

Out-of-band is a flag, not a verdict. The corruptions license a one-directional inference: damage drives charts out of band. The converse is not certified, and should not be—a generator that produces charts outside the human envelope while passing the constraint gates, coupling to the music, and retaining learnable structure is a candidate for originality, not a failure. In the novelty-plus-value framing of creativity, the typicality axes measure the magnitude of novelty while the defect-type axes and the rest of the profile supply the value judgment; the same out-of-band reading means damage when the value axes collapse and material for human review when they hold. This adjudication is itself an argument for profiles: the meaning of one axis depends on the others. The band units calibrate it empirically: held-out official charts reach up to 4.9 half-widths out on the language axis, so deviations of that order lie within the observed human envelope—Mapperatorinator’s grammar deviation (+2.1) does—whereas TaikoNation’s +12 lies three times beyond any human chart in the corpus. It is, however, a protocol rather than an exercised instrument: the difficulty-conditioning verdict of Section 3.5 is its only worked example, and instantiating the novelty arm requires human judgment data.

Beyond one game. Two families read only note times and were applied unchanged across three chart vocabularies here (timing, coupling); the token families (grammar, structure, gap) transfer with a vocabulary adapter—lane, position-bin, or direction tokens replace color tokens—plus a calibration corpus of human charts for the target game, and the constraint family re-calibrates its limits from the same corpus. The corruption operators act on (t, class) event streams and port as-is, so the certification protocol, not any single metric, is the transferable artifact: given a new game’s corpus, the same harness determines which metrics work there. Position-based games additionally need a spatial-ergonomics family (movement cost, hand allocation, rotation) that no vocabulary adaptation of the present families supplies; we leave it as the main structural extension.

Limitations. The corruptions certify damage detection, not experience prediction; validating the surviving metrics against player judgments—including the novelty-vs-damage adjudication above—remains future work. Direction is guaranteed at the population level: a minority of individual charts move toward their band under corruption, so per-chart verdicts need the profile, not one score. The suite is a coupled profile (Figure 2); attribution between timing and pattern damage is shared through tokenization, which is partly a property of the domain (mistimed notes damage patterns as played). One game corpus, of unstated genre breadth, calibrates the bands; the cross-game systems are evaluated on the game-agnostic families only; systems are run in their public inference configurations at one seed. Overlap between the test songs and the systems’ own train-

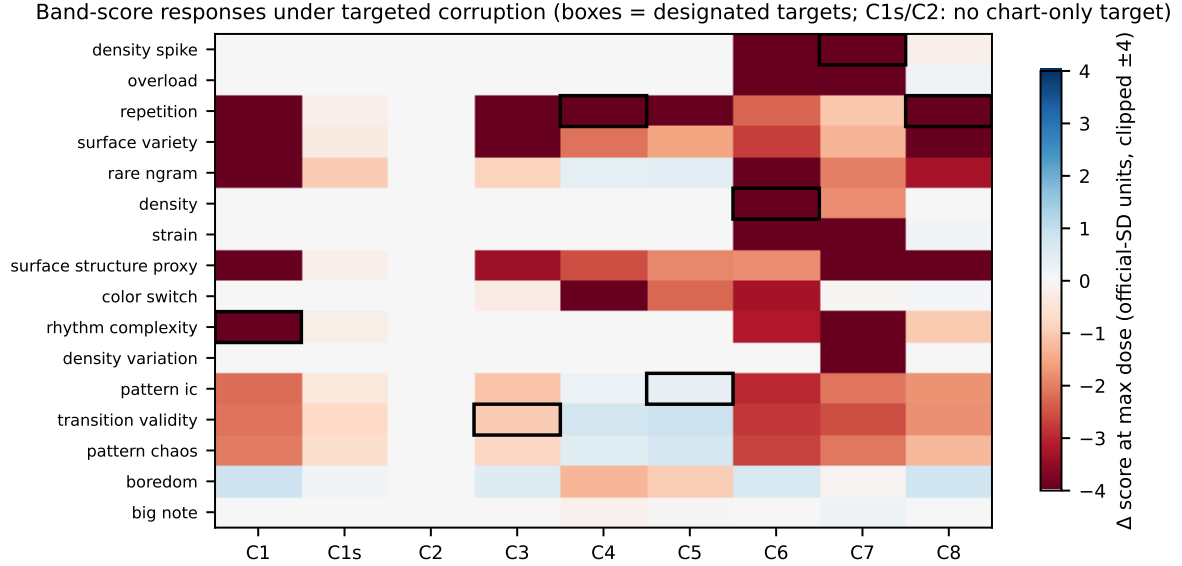


Figure 2: The operator $\times$ band-score response matrix (all 16 calibrated scores; the raw-layer matrix over all 56 quantities ships with the released records). Cells are responses at maximum dose in standard deviations of the official-chart distribution, clipped at ± 4 ; boxes = designated targets; C2 and C1s have no chart-only target). Targeted operators move their targets, but responses are *not* diagonal: timing jitter (C1) propagates into pattern scores because pattern tokens are IOI-quantized—in a rhythm game, timing damage is also pattern damage. We publish the matrix as a coupling map rather than claiming orthogonality; band-layer cells for C1s are pale because one-to-three-note damage is below band resolution, and its detection lives in the raw diagnostic layer (Figure 1), which is the division of labor the role taxonomy prescribes.

ing corpora was not controlled, and per-format conversion offsets, though bounded by the shared-path control of Section 2.5, cannot be fully excluded: the system profiles are diagnostic characterizations, not a leaderboard. The operator set spans the family taxonomy but not all damage classes: chart–audio mismatch (wrong-song coupling), salient-note deletion, and intentional off-lattice styles are untested.

5 Conclusion

For chart generation, validated evaluation is currently scarcer than new metrics. Corruption-based validation is systematic and reproducible, and it showed that the borrowed metrics did not merely underperform: they rewarded flattening, rewarded collapse, missed a global misalignment ten times the perceptual threshold, and inverted ranking direction across system regimes. Corpus-calibrated band scores over the same raw quantities, extended with a perceptually anchored timing family over a grid-reference hierarchy, pass the same corruptions—population-level monotone dose response, bounded response to the manipulation attack, exact shift recovery—and their coupling structure is published rather than hidden. Applied to five published systems, the resulting profiles replace rankings with mechanism-level diagnoses: systems with equal scores and disjoint failure axes, progress that single numbers cannot register, and difficulty-conditioning failures that only per-difficulty calibration detects. The harness, calibration artifacts, per-chart records, and evaluation package are released so that future chart-quality metrics—for this game or others—can be validated under the same protocol. [PEND-

ING: P8: artifact URL.]

Ethics Statement

The corpus consists of the community-transcribed official-course charts of Section 2.1, used under research terms; released per-chart records contain metric values, not note data, and do not reconstruct the charts. No personal data is involved. Better automated evaluation chiefly benefits players and small developers; we see no direct misuse vector beyond automated content farming, which platform moderation already addresses.

AI Usage Statement

AI assistants were used for experiment orchestration, drafting, and editing under author direction; all numbers were produced by the released code, adversarially re-derived, and verified by the authors.

References

- [1] C. Donahue, Z. C. Lipton, and J. McAuley, “Dance dance convolution,” in *Proc. of the 34th International Conference on Machine Learning (ICML)*, 2017, pp. 1039–1048.
- [2] E. Halina and M. Guzdial, “Taikonation: Patterning-focused chart generation for rhythm action games,” in *Proc. of the 16th International Conference on the Foundations of Digital Games (FDG)*, 2021.

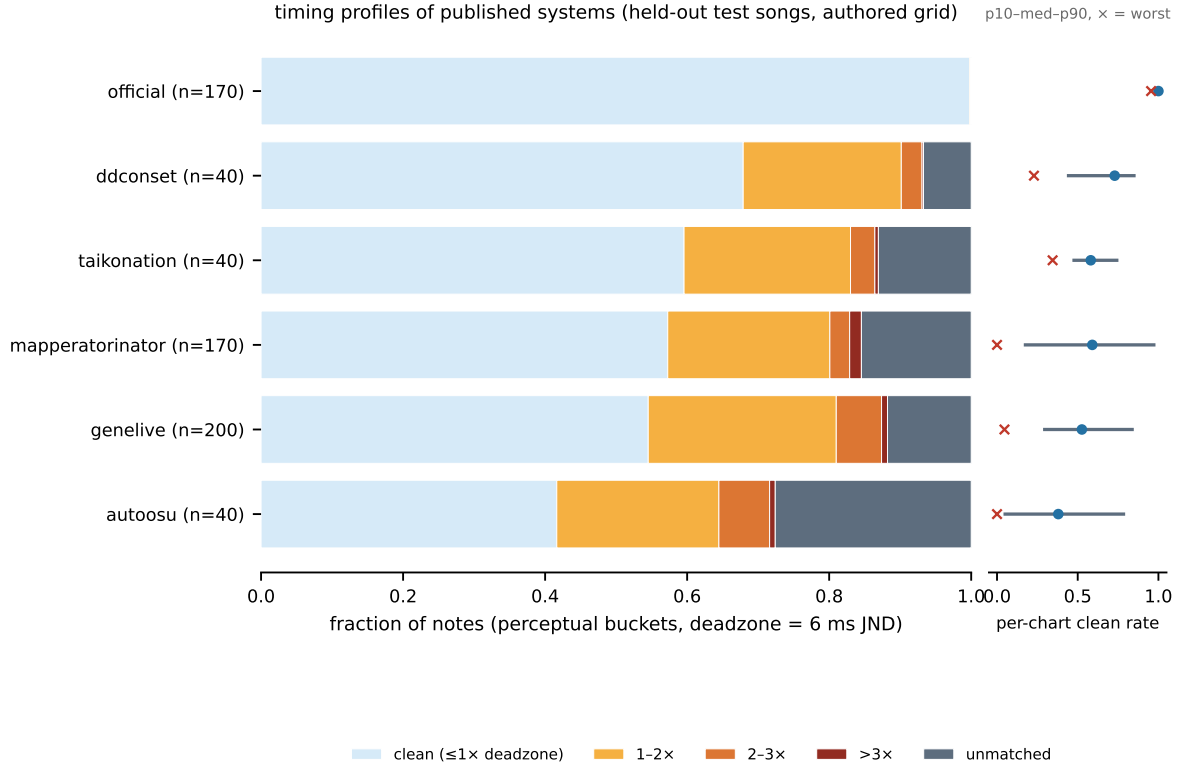


Figure 3: Timing profiles of published systems on the held-out test set (authored grid). Bars show the mean over charts of per-chart bucket fractions—clean ($\leq 1\times$ the 6 ms deadzone), tier exceedances, and unmatched—with rows ordered by the same statistic; n is the number of charts (difficulty-conditioned systems produce one chart per official course, others one per song). Right strip: per-chart clean-rate p10–median–p90 with the worst chart marked $\times$ —tail charts that corpus means hide. The official row is all-clean by construction (officials are authored on the grid that defines cleanliness) and serves as a consistency check.

- [3] A. Takada, D. Yamazaki, Y. Yoshida, N. Ganbat, T. Shimotomai, N. Yamamoto, Y. Iwasawa, and Y. Matsuo, “Genélive! generating rhythm actions in love live!” in *Proc. of the AAAI Conference on Artificial Intelligence (IAAI track)*, 2023.
- [4] J. Hanzen, E. Halina, and M. Guzdial, “Time-based chart partitioning: Improving local coherency in rhythm game chart generation,” *Proc. of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, vol. 21, no. 1, pp. 43–51, 2025.
- [5] J. Li, M. Galley, C. Brockett, J. Gao, and B. Dolan, “A diversity-promoting objective function for neural conversation models,” in *Proc. of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, 2016, pp. 110–119.
- [6] Y. Zhu, S. Lu, L. Zheng, J. Guo, W. Zhang, J. Wang, and Y. Yu, “Texygen: A benchmarking platform for text generation models,” in *Proc. of the 41st International ACM SIGIR Conference on Research & Development in Information Retrieval*, 2018, pp. 1097–1100.
- [7] L.-C. Yang and A. Lerch, “On the evaluation of generative models in music,” *Neural Computing and Applications*, vol. 32, no. 9, pp. 4773–4784, 2020.
- [8] H.-W. Dong, K. Chen, J. McAuley, and T. Berg-Kirkpatrick, “MusPy: A toolkit for symbolic music generation,” in *Proc. of the 21st International Society for Music Information Retrieval Conference (ISMIR)*, 2020.
- [9] S.-L. Wu and Y.-H. Yang, “The jazz transformer on the front line: Exploring the shortcomings of ai-composed music through quantitative measures,” in *Proc. of the 21st International Society for Music Information Retrieval Conference (ISMIR)*, 2020.
- [10] L. Theis, A. van den Oord, and M. Bethge, “A note on the evaluation of generative models,” in *Proc. of the 4th International Conference on Learning Representations (ICLR)*, 2016.
- [11] C. Raffel, B. McFee, E. J. Humphrey, J. Salamon, O. Nieto, D. Liang, and D. P. W. Ellis, “mir_eval: A transparent implementation of common MIR metrics,” in *Proc. of the 15th International Society for Music Information Retrieval Conference (ISMIR)*, 2014.

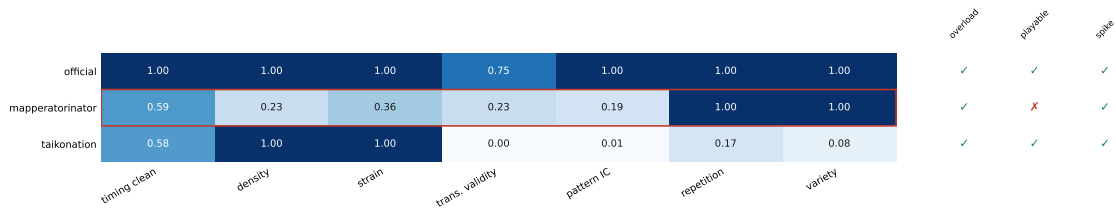


Figure 4: Ranker-axis profiles of the three taiko-vocabulary systems (cells are 0–1 scores: timing clean rate and calibrated band scores; a score of 1.00 means in-band, and small values are readable as band distance under the rational falloff—e.g. TaikoNation’s repetition 0.17 is 3.3 half-widths below band while its pattern-IC 0.01 is 12 above). The two generated systems deviate on complementary axes: TaikoNation holds density and strain but collapses on every pattern axis; Mapperatorinator holds repetition and variety but deviates on density and strain. Right glyphs: constraint gates, reported separately because gates veto rather than average (pass = median band score ≥ 0.5 ; a failed gate outlines the row); Mapperatorinator trips the playability gate. Cross-vocabulary systems appear in Figure 3 and the text. The official row reads 0.75 on transition validity because a fixed training-corpus LM assigns nonzero unseen-transition rates to new songs; that value, not 1.0, is the human reference level for that column.

- [12] A. Friberg and J. Sundberg, “Time discrimination in a monotonic, isochronous sequence,” *Journal of the Acoustical Society of America*, vol. 98, no. 5, pp. 2524–2531, 1995.
- [13] J. London, *Hearing in Time: Psychological Aspects of Musical Meter*, 2nd ed. Oxford University Press, 2012.
- [14] OliBomby, “Mapperatorinator: multi-gamemode osu! beatmap generation (v32),” <https://github.com/OliBomby/Mapperatorinator>, 2024, open-source system; weights on Hugging Face.
- [15] AutoOsu contributors, “Autoosu: difficulty-targeted osu! beatmap generation,” GitHub, 2023, open-source system [PENDING: P7: verify repo/author].